

PROJECT REPORT

SCRIPT-CONTROLLED ACL – RESTRICTED ACCESS BASED ON FIELD VALUE

1. INTRODUCTION

Project Overview

The project, **Script-Controlled ACL – Restricted Access Based on Field Value**, is developed using ServiceNow to control access to records dynamically. The system uses Access Control Lists (ACLs) and script-based conditions to determine whether a user is authorized to access a record based on the value of a specific field.

The system evaluates the field value of a record and checks the user's role or authorization before allowing access. If the user satisfies the configured access condition, access is granted. Otherwise, access to the restricted record is denied.

Purpose

The purpose of the system is to provide **dynamic and field-based record security**, reduce manual access configuration, prevent unauthorized access, improve data protection, and provide flexible access control based on business requirements.

2. IDEATION PHASE

Problem Statement

In ServiceNow applications, providing the same level of access to all users may expose sensitive or restricted records. Static role-based permissions may not always be sufficient when access needs to depend on the value of a particular field.

For example, records may contain different departments, locations, priorities, classifications, or confidentiality levels. Users should only be able to access records for which they have the required authorization.

Therefore, a script-controlled ACL solution is required to dynamically restrict record access based on field values.

Empathy Map Canvas

Section	Content
Think & Feel	Concerned about unauthorized users accessing restricted records.

Hear	Requests from users and administrators for controlled access to specific records.
See	Records containing different field values that require different access permissions.
Say & Do	Configure roles, define access conditions, test ACL rules, and manage user permissions.
Pain	Static access rules, manual configuration, unauthorized access risks, and complex permission management.
Gain	Dynamic access control, improved security, reduced manual work, and controlled record visibility.

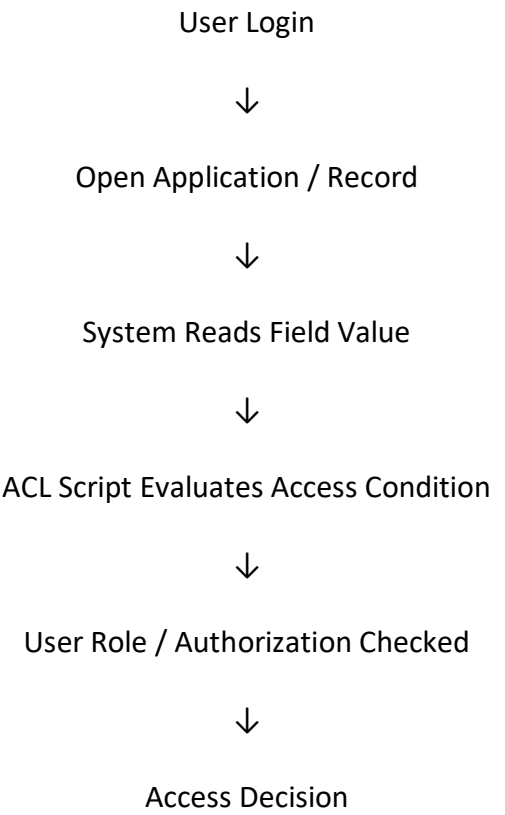
Brainstorming

The proposed ideas included:

- Field-value-based access control.
- Script-controlled ACL.
- Role-based access validation.
- Dynamic record-level security.
- Unauthorized access restriction.
- Automated access validation.
- Easy maintenance and future scalability.

3. REQUIREMENT ANALYSIS

Customer Journey Map





Authorized → Allow Record Access

Unauthorized → Restrict Record Access



Access Result Displayed

Solution Requirement

The system must:

- Restrict record access based on a specific field value.
- Use ServiceNow ACL mechanisms for access control.
- Execute a script to evaluate access conditions dynamically.
- Validate the user's role or authorization.
- Allow authorized users to access permitted records.
- Restrict unauthorized users from accessing restricted records.
- Support different access conditions for different field values.
- Handle invalid or missing field values securely.
- Support easy maintenance and future scalability.

Data Flow Diagram

User



ServiceNow Application



ServiceNow Record



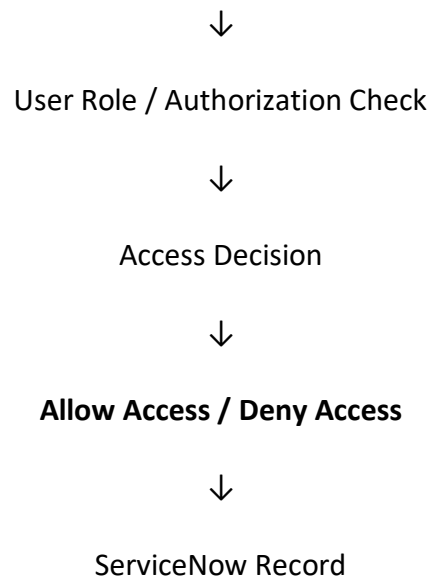
Field Value



ACL Rule



Script Evaluation



Technology Stack

Component	Technology
Platform	ServiceNow
Access Control	ServiceNow ACL
Business Logic	ACL Script
Database	ServiceNow Tables
User Management	ServiceNow Users and Roles
Authorization	ServiceNow Roles and Permissions
Scripting	JavaScript
Testing	ServiceNow ACL Testing / User Access Testing

4. PROJECT DESIGN

Problem Solution Fit

The solution directly addresses the problem of unauthorized record access by introducing field-value-based access control. Instead of depending only on general user roles, the system evaluates the value of a specific field and dynamically determines whether the current user should be allowed to access the record.

This provides more granular control over records and helps protect restricted information.

Proposed Solution

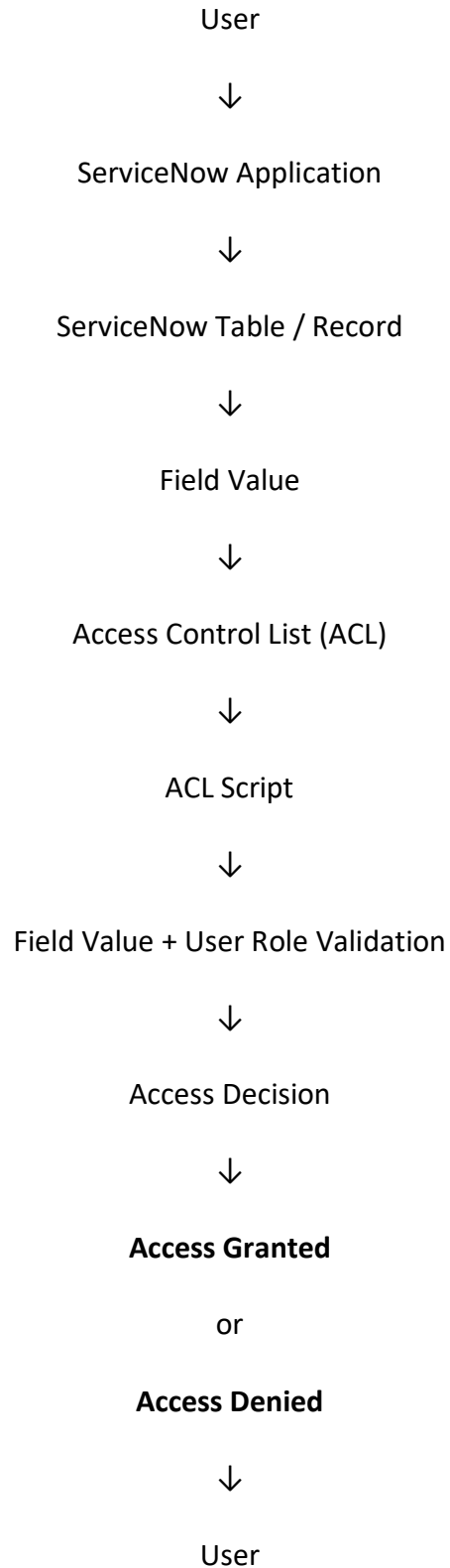
The proposed solution uses a **ServiceNow ACL with a script-controlled condition**. When a user attempts to access a record, the ACL evaluates the relevant field value and checks the user's authorization.

Based on the configured conditions:

- Authorized users are granted access.

- Unauthorized users are denied access.
- Different field values can have different access requirements.
- Changes in field values can result in different access decisions.

Solution Architecture



PROJECT PLANNING & SCHEDULING

Project Planning

The project is divided into **5 Epics and 12 User Stories**, with a total of **31 Story Points**.

Epic	Description	Assigned To
EPIC-1	ACL Table & Field Configuration	Vashandamil ilakya D, Dhivya Bharathi V, Hariharan M
EPIC-2	Field-Based Access Rule	Maheshwari K, Praveen kumar D
EPIC-3	Script-Controlled ACL	Vashandamil ilakya D, Dhivya Bharathi V, Hariharan M
EPIC-4	Access Validation & Testing	Maheshwari K, Praveen kumar D
EPIC-5	Deployment & Documentation	Vashandamil ilakya D

Project Start Date: 01/10/2026

Project End Date: 03/10/2026

Team Lead: Vashandamil ilakya D

Team Members: Vashandamil ilakya D, Dhivya Bharathi V, Hariharan M, Maheshwari K, Praveen kumar D

6. FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing

The system was tested to verify that the script-controlled ACL correctly evaluates field values and user authorization before allowing access to records.

Test Case	Input	Expected Output	Result
TC-01	Authorized user + permitted field value	Record access granted	Pass
TC-02	Unauthorized user + restricted field value	Record access denied	Pass
TC-03	Authorized role + restricted record	Access based on configured ACL condition	Pass
TC-04	Unauthorized role + permitted field value	Access denied	Pass

TC-05	Missing/invalid field value	Restricted access / safe denial	Pass
TC-06	Field value changed	ACL evaluates the updated value	Pass

The tests confirm that the system evaluates the configured field value and user authorization before granting record access.

7. RESULTS

Output Screenshot:

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☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
	EEE			Aileen Mottern		Abel Tuter	STU111
	ECE			Alfonso Griglen		Adela Cervantsz	STU222
	CSE			Andrew Och		Allan Schwantd	STU333

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Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Aileen Mottern		Abel Tuter	STU111

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides field-value-based record access control.
- Restricts unauthorized users from accessing protected records.
- Provides more granular security than basic role-based access.
- Dynamically evaluates access conditions.
- Supports different access rules for different field values.
- Reduces manual access management.
- Improves protection of sensitive records.
- Can be extended with additional conditions.
- Uses ServiceNow ACL and scripting capabilities.
- Supports centralized access-control management.

Disadvantages

- Access depends on correctly configured ACL conditions.
- Incorrect scripts may result in unintended access restrictions.
- Changes to field values or requirements may require ACL updates.
- Requires appropriate ServiceNow roles and permissions for configuration.
- Complex access requirements may require more complex scripting.
- Testing is required whenever ACL conditions are modified.

9. CONCLUSION

The **Script-Controlled ACL – Restricted Access Based on Field Value** successfully addresses the requirement for dynamic and granular record-level access control in ServiceNow.

The system evaluates the field value of a record together with the user's authorization and determines whether access should be granted or denied. This approach provides controlled access to records and reduces the possibility of unauthorized users viewing restricted information.

The implemented solution provides a structured and flexible approach for managing field-based access restrictions and can be extended to support additional access-control requirements.

10. FUTURE SCOPE

The system can be further enhanced by:

- Adding multiple field-based access conditions.
- Supporting more complex combinations of fields.
- Adding department-based access control.
- Adding location-based access restrictions.
- Supporting dynamic role assignment.
- Adding an administrator dashboard for ACL monitoring.
- Adding detailed access-denied and access-granted logs.
- Integrating automated access request and approval workflows.
- Adding notification when unauthorized access is attempted.
- Supporting more advanced security rules.
- Integrating additional ServiceNow applications and tables.
- Adding reporting and analytics for access-control activities.
- Extending the solution for larger enterprise access-control requirements.

Demo link: https://drive.google.com/file/d/1KwFxJdds28DTTtjTiS-_yXFS7UsAltup/view?usp=sharing

Github Link: